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Sample Paper: Data Science

Total Marks: 100

Time Allowed: 3 Hours

Section A: Python Programming (20 Marks)

1. **Short Questions** (2 marks each):
 - a. What is a lambda function in Python? Write a lambda function to compute the square of a number.
 - b. How do you read a CSV file in Python? Mention the library and method used.
 - c. Explain the difference between `iloc` and `loc` in Pandas.
 2. **Coding Question** (8 marks):

Write a Python program to read a CSV file containing student scores, calculate the average score for each student, and save the results to a new CSV file.
-

Section B: Data Manipulation and Cleaning (20 Marks)

3. **Short Questions** (2 marks each):
 - a. What is a Pandas DataFrame? How is it different from a Series?
 - b. Explain the purpose of the `groupby` function in Pandas.
 - c. What is the difference between `merge` and `join` in Pandas?
 4. **Practical Question** (8 marks):

You are given a dataset with missing values in the "Age" column. Write a Python script to:

 - Fill missing values with the mean age.
 - Remove rows where more than 50% of the data is missing.
-

Section C: Inferential Statistics (20 Marks)

5. **Short Questions** (3 marks each):
 - a. Define p-value and its significance in hypothesis testing.
 - b. Explain the difference between descriptive and inferential statistics.
 6. **Numerical Questions** (7 marks each):
 - a. A survey was conducted to determine the average age of employees in a company. A sample of 30 employees has an average age of 35 years with a standard deviation of 5 years. Calculate the 95% confidence interval for the population mean.
 - b. A company claims that their new product reduces defects by 10%. A sample of 50 units showed a mean defect reduction of 8% with a standard deviation of 2%. Perform a hypothesis test (using a 5% significance level) to verify this claim.
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Section D: Machine Learning (40 Marks)

7. **Conceptual Questions** (2 marks each):
 - a. Differentiate between supervised and unsupervised learning.
 - b. What is the purpose of cross-validation in machine learning?
 - c. Explain the concept of overfitting and how it can be avoided.
8. **Numerical Questions** (10 marks each):
 - a. A linear regression model is trained to predict house prices. Given the following data:
 - Predicted prices: [150k, 200k, 250k, 300k]
 - Actual prices: [140k, 210k, 240k, 310k]
 Calculate the Mean Squared Error (MSE).
 - b. Perform K-Means clustering on the dataset:

css
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 Points = [(1,1), (2,1), (4,3), (5,4)]
 Initial Centroids = [(1,1), (5,4)]
9. Show the first iteration of the clustering process.
10. **Case Study** (10 marks):

You are given a dataset with the following columns: "Age," "Income," "Spending Score."

 - Perform dimensionality reduction using PCA.
 - Explain how PCA can help in machine learning tasks.

Sample Paper: Advanced Data Science

Total Marks: 100

Time Allowed: 3 Hours

Section A: Python Programming (20 Marks)

1. **Short Questions** (3 marks each):
 - a. Write a Python function using a lambda expression to filter a list of numbers that are divisible by 3.
 - b. Explain how `itertools` can be used for efficient iteration in Python. Provide an example.
2. **Coding Challenge** (14 marks):

Write a Python program to:

 - Read a large CSV file (500MB+) in chunks using Pandas.
 - Calculate the mean, median, and standard deviation for a numeric column.
 - Save these statistics into a summary CSV file.

Section B: Data Manipulation and Cleaning (20 Marks)

3. **Conceptual Questions** (4 marks each):
 - a. Explain the difference between `apply()` and `map()` in Pandas. Which one is more flexible and why?
 - b. Discuss the trade-offs of using the `merge()` function versus concatenation in Pandas when dealing with large datasets.
4. **Complex Data Cleaning Task** (12 marks):

You are given a dataset with the following issues:

 - Duplicate rows.
 - Missing values in multiple columns.
 - Categorical data not encoded numerically.
5. Write a Python script to:
 - Remove duplicates.
 - Replace missing numeric values with the median.
 - Encode categorical data using one-hot encoding.

Section C: Inferential Statistics (20 Marks)

5. **Advanced Numerical Questions** (10 marks each):
 - a. A sample of 100 people has a mean weight of 70 kg with a standard deviation of 10 kg. Conduct a hypothesis test to determine if the population mean weight is significantly different from 68 kg at a 99% confidence level.
 - b. A company wants to test if a new drug increases recovery rates. They collected the following data:
 - Group A (Control, n=50): Mean recovery rate = 60%, SD = 8%.
 - Group B (Treatment, n=50): Mean recovery rate = 65%, SD = 7%.Perform a two-sample t-test (5% significance level) to conclude if the treatment is effective.

Section D: Machine Learning (40 Marks)

6. **Conceptual Questions** (5 marks each):
 - a. Explain how Principal Component Analysis (PCA) works. Why is it important to standardize data before applying PCA?
 - b. Discuss the role of hyperparameter tuning in machine learning. Give examples of hyperparameters for **Decision Trees** and **Support Vector Machines**.
7. **Numerical Problems** (10 marks each):
 - a. A decision tree is trained on a dataset. Given the following split:
 - Parent Node: 80 samples, Gini impurity = 0.5.
 - Left Child: 50 samples (80% of class A, 20% of class B).
 - Right Child: 30 samples (30% of class A, 70% of class B).
8. Calculate the information gain of the split.
 - b. Perform k-fold cross-validation (k=5) for the following data:
 - Dataset: 100 samples
 - Model: Logistic Regression with accuracy values for each fold as [0.85, 0.89, 0.82, 0.88, 0.87].Calculate the mean accuracy and standard deviation.
9. **Case Study: Real-World Application** (10 marks):

A company wants to cluster customers based on their purchasing behavior.

 - Dataset: "Annual Income" and "Spending Score."
 - Tasks:
 - Apply K-Means clustering (k=3).
 - Evaluate the clustering using silhouette score.
 - Suggest improvements if the silhouette score is low.

Sample Paper: Advanced Data Science

Total Marks: 100

Time Allowed: 3 Hours

Section A: Python Programming (20 Marks)

1. **Short Questions** (3 marks each):
 - a. What is the difference between deep copy and shallow copy in Python? Provide an example.

b. How can you use Python's `collections` module to count the frequency of elements in a list?

2. **Coding Challenge** (14 marks):

Write a Python program to:

- Read a JSON file containing nested data.
- Extract specific keys and save them into a new DataFrame.
- Export the DataFrame as a CSV file, ensuring the columns are properly labeled.

Section B: Data Manipulation and Cleaning (20 Marks)

3. **Conceptual Questions** (4 marks each):

a. What is the difference between `.pivot()` and `.pivot_table()` in Pandas? Provide a scenario where each should be used.

b. Explain the purpose of using multi-indexing in Pandas. How does it improve data analysis for hierarchical data?

4. **Complex Data Manipulation Task** (12 marks):

You are given a dataset with:

- A column "Transaction Date" in string format.
- A column "Sales" with outliers.
- A column "Product Category" with inconsistent capitalization.

5. Write a Python script to:

- Convert "Transaction Date" to a datetime object and extract the year and month.
- Handle outliers in the "Sales" column using the IQR method.
- Standardize the capitalization for "Product Category."

Section C: Inferential Statistics (20 Marks)

5. **Advanced Numerical Questions** (10 marks each):

a. A company claims that their new process reduces production time. A sample of 40 production cycles has a mean time of 50 hours with a standard deviation of 5 hours. Conduct a one-sample z-test to determine if the process time is less than the previous mean of 52 hours (use a 5% significance level).

b. Two marketing strategies were tested on different groups of customers.

- Group A (n=30): Mean sales = \$10,000, SD = \$1,500.
- Group B (n=30): Mean sales = \$11,500, SD = \$2,000.

Perform a two-tailed independent t-test (5% significance level) to conclude if there is a significant difference between the strategies.

Section D: Machine Learning (40 Marks)

6. **Conceptual Questions** (5 marks each):
- What is the Curse of Dimensionality? How does it affect machine learning models?
 - Explain the differences between Bagging and Boosting with examples of algorithms for each.
7. **Numerical Problems** (10 marks each):
- A dataset has the following class distribution:
 - Class A: 900 samples
 - Class B: 100 samples

A Random Forest model predicts the following confusion matrix for a test dataset:

yaml

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```
Predicted: A      B
Actual A: 850    50
Actual B: 60     40
```

8. Calculate:
- Accuracy
 - Precision for Class B
 - Recall for Class B
9. b. Perform one iteration of the Gradient Descent algorithm:
- Hypothesis: $h(x) = \theta_0 + \theta_1 x$
 - Initial $\theta_0 = 1$, $\theta_1 = 2$, Learning rate (α) = 0.1
 - Dataset: $[(x=1, y=4), (x=2, y=7)]$
10. Calculate the updated values of θ_0 and θ_1 .
11. **Case Study: Real-World Application** (10 marks):
- A retailer wants to predict customer churn. You are given the following columns:
- "Customer Age," "Annual Income," "Tenure," "Churn (Yes/No)."
12. Tasks:
- Train a Logistic Regression model to predict churn.
 - Evaluate the model using the confusion matrix, precision, and recall.
 - Suggest improvements if the model performs poorly.

Sample Paper: Advanced Data Science

Total Marks: 100

Time Allowed: 3 Hours

Section A: Python Programming (20 Marks)

1. **Short Questions** (3 marks each):
 - a. Explain the use of Python's `zip()` function with an example.
 - b. How does Python's `argparse` module assist in writing command-line tools?
 2. **Coding Challenge** (14 marks):

Write a Python program to:

 - Read a dataset from an Excel file with multiple sheets.
 - Combine the data from all sheets into a single DataFrame.
 - Group the data by a categorical column and calculate the median for a numeric column.
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Section B: Data Manipulation and Cleaning (20 Marks)

3. **Conceptual Questions** (4 marks each):
 - a. Discuss the difference between `.sort_values()` and `.rank()` in Pandas. Provide use cases for each.
 - b. Explain the role of the `query()` method in Pandas. How does it improve readability and performance?
 4. **Data Manipulation Task** (12 marks):

You are provided a dataset with the following columns:

 - "Order Date" (string format), "Region," "Sales," and "Profit Margin."
 5. Write a Python script to:
 - Convert "Order Date" to datetime format and extract the year.
 - Normalize the "Sales" column to a 0-1 scale.
 - Identify regions where the average profit margin is below 10%.
-

Section C: Inferential Statistics (20 Marks)

5. **Advanced Numerical Questions** (10 marks each):
 - a. A sample of 25 students was surveyed to determine the average number of hours they study per week. The sample mean was 15 hours, and the standard deviation was 3 hours. Construct a 99% confidence interval for the population mean.
 - b. A fitness tracker company claims its device accurately measures steps. A sample of 40 devices was tested, showing a mean step count error of 5 steps with a standard deviation of 2 steps. At a 5% significance level, test if the error is significantly different from the claimed error of 4 steps.
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Section D: Machine Learning (40 Marks)

6. **Conceptual Questions** (5 marks each):
 - a. Explain the differences between Grid Search and Random Search for

hyperparameter tuning. Discuss the advantages and limitations of each.
 b. How does the choice of distance metric (e.g., Euclidean vs. Manhattan) affect the performance of K-Nearest Neighbors (KNN)?

7. **Numerical Problems** (10 marks each):

a. A logistic regression model predicts the following probabilities for a binary classification task:

- Predicted probabilities: [0.8, 0.3, 0.6, 0.4, 0.7]
- Actual labels: [1, 0, 1, 0, 1]

Calculate the log-loss value for these predictions.

b. A dataset has the following points for K-Means clustering (k=2):

yaml

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Points: [(2, 3), (6, 6), (3, 4), (8, 8)]

Initial centroids: [(2, 3), (8, 8)]

8. Perform the first iteration of the clustering process and update the centroids.

9. **Case Study: Real-World Application** (10 marks):

A financial institution wants to detect fraudulent transactions. The dataset includes columns like "Transaction Amount," "Merchant Type," "Time of Transaction," and "Fraudulent (Yes/No)."

Tasks:

- Train a Random Forest model to classify transactions as fraudulent or non-fraudulent.
- Explain how feature importance can be used to identify patterns in fraudulent transactions.
- Suggest additional features that could improve the model's performance.

Sample Paper: Advanced Data Science

Total Marks: 100

Time Allowed: 3 Hours

Section A: Python Programming (20 Marks)

1. **Short Questions** (3 marks each):

- a. What is the difference between `is` and `==` in Python? Provide examples.
- b. Explain the use of Python's `with` statement. How does it enhance code readability and reliability?

2. Coding Challenge (14 marks):

Write a Python program to:

- Read a dataset from a CSV file.
- Filter rows where a numeric column exceeds its 90th percentile value.
- Save the filtered rows into a new CSV file.

Section B: Data Manipulation and Cleaning (20 Marks)

3. Conceptual Questions (4 marks each):

- a. Discuss the trade-offs between using `.apply()` and vectorized operations in Pandas for data transformation.
- b. What is the purpose of using `astype()` in Pandas? Provide examples where type conversion is necessary for data analysis.

4. Data Manipulation Task (12 marks):

You are provided with a dataset containing:

- "Employee ID," "Joining Date," "Department," and "Annual Salary."

5. Write a Python script to:

- Extract employees who joined before 2015.
- Calculate the average salary for each department.
- Identify the department with the highest average salary.

Section C: Inferential Statistics (20 Marks)

5. Advanced Numerical Questions (10 marks each):

a. A company tested two different marketing campaigns. The results are as follows:

- Campaign A: Sample Size = 50, Mean Revenue = \$12,000, SD = \$2,500
- Campaign B: Sample Size = 60, Mean Revenue = \$13,500, SD = \$2,800

Conduct a two-sample t-test (5% significance level) to determine which campaign is more effective.

b. A dataset shows the scores of students in two subjects: Math and Science. The scores are as follows:

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Math: [75, 80, 85, 90, 95]

Science: [70, 78, 83, 88, 92]

6. Perform a Pearson correlation analysis to determine the strength of the relationship between the two subjects.

Section D: Machine Learning (40 Marks)

6. **Conceptual Questions** (5 marks each):
- What are the key differences between classification and regression tasks? Provide examples for each.
 - Explain how the learning rate affects the convergence of Gradient Descent. What happens if the learning rate is too high or too low?
7. **Numerical Problems** (10 marks each):
- A decision tree splits data as follows:
 - Parent Node: 100 samples, Gini Impurity = 0.5
 - Left Child: 60 samples (40% of class A, 60% of class B)
 - Right Child: 40 samples (70% of class A, 30% of class B)
 - Calculate the information gain of the split.
 - A dataset with 100 samples is split into training (80%) and testing (20%). A machine learning model trained on the dataset achieves the following results on the test set:
 - True Positives: 16
 - True Negatives: 18
 - False Positives: 2
 - False Negatives: 4
 - Calculate the following metrics:
 - Accuracy
 - Precision
 - Recall
 - F1 Score
10. **Case Study: Real-World Application** (10 marks):
- A ride-sharing company wants to predict demand during peak hours. The dataset includes columns like "Hour of Day," "Weather Condition," "Day of Week," and "Number of Rides."
- Tasks:
- Train a Linear Regression model to predict the number of rides.
 - Evaluate the model using Mean Squared Error (MSE).
 - Suggest additional features to improve the model's accuracy.

Total Marks: 100

Time Allowed: 3 Hours

Section A: Python Programming (20 Marks)

1. **Short Questions** (3 marks each):
 - a. Discuss the time complexity of the following operations in Python:
 - Accessing an element in a list.
 - Inserting an element into a dictionary.
 - b. What is the purpose of Python's `functools.lru_cache`? Provide a use case for it.
 2. **Complex Coding Challenge** (14 marks):

Write a Python program to:

 - Read a large dataset from a SQL database using Python's `sqlalchemy`.
 - Perform data aggregation, grouping the data by a categorical column and calculating the sum, mean, and count for a numeric column.
 - Save the results to a new table in the database.
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Section B: Data Manipulation and Cleaning (20 Marks)

3. **Conceptual Questions** (4 marks each):
 - a. Explain how Pandas handles missing data using `isna()` and `fillna()`. What are the computational trade-offs of using these methods on large datasets?
 - b. Compare the performance of Pandas and Dask for data manipulation on datasets larger than memory.
 4. **Advanced Data Cleaning Task** (12 marks):

You are given a dataset with:

 - A column "Timestamp" in Unix format.
 - Duplicate entries with slight differences in numeric columns due to rounding errors.
 5. Write a Python script to:
 - Convert the "Timestamp" column to a human-readable datetime format.
 - Identify and remove duplicates based on a 5% tolerance for numeric column differences.
-

Section C: Inferential Statistics (20 Marks)

5. **Advanced Numerical Questions** (10 marks each):
 - a. A researcher wants to study the effect of a new fertilizer on plant growth. Two groups were tested:
 - Group A (Control): Mean height = 15 cm, SD = 2 cm, n = 30.
 - Group B (Treated): Mean height = 18 cm, SD = 2.5 cm, n = 30.

6. Conduct a two-tailed hypothesis test at a 1% significance level to determine if the fertilizer has a significant effect.
 - b. A company surveyed customers to determine if income affects spending habits. The results are as follows:
 - Income (\$): [50k, 60k, 70k, 80k, 90k]
 - Spending (\$): [5k, 6.2k, 7.1k, 8.3k, 9.5k]
 7. Perform a linear regression analysis to determine the relationship between income and spending. Calculate the slope, intercept, and R-squared value.
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Section D: Machine Learning (40 Marks)

6. **Conceptual Questions** (5 marks each):
 - a. Explain the difference between Ridge and Lasso regression. When would you choose one over the other?
 - b. What is the difference between a Generative and Discriminative model? Give examples of each.
7. **Numerical Problems** (10 marks each):
 - a. A Support Vector Machine (SVM) is trained with the following data:
 - Class A: $[(-1, -1), (-2, -1), (-1, -2)]$
 - Class B: $[(1, 1), (2, 1), (1, 2)]$
 - b. Derive the decision boundary equation using the SVM optimization problem.
8. Derive the decision boundary equation using the SVM optimization problem.
 - b. A dataset with 500 samples is used to train a Random Forest model with 100 trees. Each tree uses 60% of the data for training and 40% for testing (OOB samples). Explain how OOB error is calculated and demonstrate it with a hypothetical example.
9. **Case Study: Real-World Application** (10 marks):

A healthcare company wants to predict patient readmission to hospitals. The dataset includes the following columns:

 - "Age," "Length of Stay," "Diagnosis," "Readmitted (Yes/No)."
10. Tasks:
 - Use a Decision Tree to predict readmissions.
 - Perform feature importance analysis to identify the most important predictors.
 - Suggest actions to improve the model's performance if it shows overfitting.

Sample Paper: Advanced Data Science (Theory-Focused)

Total Marks: 100

Time Allowed: 3 Hours

Section A: Python Programming and Basics (20 Marks)

1. **Short Questions** (5 marks each):
 - a. Explain the difference between mutable and immutable objects in Python with examples. How does this difference affect performance?
 - b. What are Python decorators? Illustrate their use with an example.
 2. **Conceptual Question** (10 marks):

Describe the lifecycle of a Python program from script execution to memory management, focusing on the role of the Global Interpreter Lock (GIL).
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Section B: Data Manipulation and Cleaning (20 Marks)

3. **Short Questions** (5 marks each):
 - a. What is the difference between wide and long data formats? Provide examples of when each format is preferred.
 - b. Explain the concept of data normalization and standardization. When would you choose one over the other?
 4. **Conceptual Question** (10 marks):

Discuss the challenges of handling missing data in large datasets. Compare three methods for dealing with missing values, highlighting the advantages and limitations of each.
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Section C: Inferential Statistics (20 Marks)

5. **Short Questions** (5 marks each):
 - a. What assumptions are required to perform a parametric statistical test? How do these differ from non-parametric tests?
 - b. Define and differentiate between Type I and Type II errors in hypothesis testing. Provide examples of their implications in real-world scenarios.
 6. **Conceptual Question** (10 marks):

Explain the Central Limit Theorem (CLT) and its significance in inferential statistics. How does the CLT relate to the construction of confidence intervals?
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Section D: Machine Learning (40 Marks)

7. **Conceptual Questions** (5 marks each):
 - a. What is feature engineering, and why is it critical in machine learning?
 - b. Explain the concept of bias-variance tradeoff. How does it affect model performance?
 - c. Describe the differences between Bagging and Boosting. Provide examples of algorithms that use these techniques.
 - d. What is regularization in machine learning? Compare L1 and L2 regularization in terms of their impact on model coefficients.

8. **Case Study Questions** (10 marks each):
 - a. A company has a dataset with highly imbalanced classes (95% in one class and 5% in the other). Discuss three approaches to handle class imbalance and explain when each approach is most effective.
 - b. A dataset has high dimensionality (10,000+ features). Explain three strategies to reduce dimensionality, detailing the advantages and disadvantages of each.
9. **Essay Question** (10 marks):

Machine learning models are often described as "black boxes." Discuss the importance of model interpretability in critical applications such as healthcare and finance. How do techniques like SHAP and LIME help address this issue?

Sample Paper: Data Science Fundamentals (Theory-Focused)

Total Marks: 100

Time Allowed: 3 Hours

Section A: Python Programming (20 Marks)

1. **Short Questions** (5 marks each):
 - a. What is the purpose of the `len()` function in Python? Provide an example of its use.
 - b. Explain the difference between a list and a tuple in Python.
2. **Conceptual Question** (10 marks):

What is a Python function? Write a simple example to explain how arguments are passed to a function.

Section B: Data Manipulation and Cleaning (20 Marks)

3. **Short Questions** (5 marks each):
 - a. What is the difference between `.iloc[]` and `.loc[]` in Pandas?
 - b. How does the `dropna()` function in Pandas handle missing values?
4. **Conceptual Question** (10 marks):

Why is data cleaning important in data science? Describe three common data cleaning tasks with examples.

Section C: Inferential Statistics (20 Marks)

5. **Short Questions** (5 marks each):
 - a. Define population and sample in the context of statistics.
 - b. What is a p-value, and how is it used in hypothesis testing?
 6. **Conceptual Question** (10 marks):

Explain the difference between descriptive and inferential statistics. Provide examples of when each is used.
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Section D: Machine Learning (40 Marks)

7. **Short Questions** (5 marks each):
 - a. What is supervised learning? Provide two examples of supervised learning algorithms.
 - b. Explain the role of a training dataset and a test dataset in machine learning.
 - c. What is overfitting, and how can it be prevented?
 - d. Define classification and regression. How are they different?
8. **Conceptual Questions** (10 marks each):
 - a. What is the purpose of a confusion matrix? Briefly explain its components (True Positive, False Positive, etc.).
 - b. Explain the K-Nearest Neighbors (KNN) algorithm. How does it classify data points?
9. **Essay Question** (10 marks):

Discuss the importance of feature selection in machine learning. How does it impact model performance?